

Instructions to students:

- 1) Illustrate the answers with neat sketches, diagram etc. wherever necessary.
- 2) Necessary data is given in the respective questions. If such data is not given it means that the knowledge of part is a part of examination.
- 3) If some part or parameter is noticed to be missing, appropriate data may be assumed and should be mentioned clearly.
- 4) Attempt any one out of Que. no. 1 and Que. No. 2
- 5) Attempt any one out of Que. no. 3 and Que. No. 4
- 6) Attempt any one out of Que. no. 5 and Que. No. 6
- 7) Attempt any one out of Que. no. 7 and Que. No. 8
- 8) Attempt any one out of remaining questions

Q.I Explain the term: Radiation resistance, Effective Area, Effective length and radiation pattern of antenna. What is the relation between gain and effective area of the antenna. Define radiation intensity. What is the relation between radiation intensity and total power radiated by an antenna.

Q.II Show that for vertical incidence of radio waves on ionosphere, the critical frequency f_c is related to the maximum electron density N_m by the equation $f_c^2 = \frac{N_m e^2}{4\pi^2 \epsilon_0 m}$, where e , ϵ_0 and m have usual significance. Express N_m in terms of f_c and a constant factor utilizing the usual values of e , ϵ_0 and m

Q. III Attempt the following

1. Define the term "Gain", "Directivity" and "Effective length" of an antenna
2. Define the term beam width of an antenna and show that its directivity is given by $D = \frac{41257}{\theta_E^\circ \theta_H^\circ}$, where θ_E° and θ_H° are half power beamwidth in E and H respectively.

Q. IV. What is Yagi antenna? Derive the necessary equation related to this. Explain its construction and properties with special reference to the directivity and bandwidth

- Q. V Design and Describe a Rhombic antenna, for the relative field strength in the vertical plane passing through the rhombic axis, derive the maximum E design formula for the rhombic design.
- Q. VI. Calculate the transmission power distance for an isotropic transmission that utilize a layer of height 200km. The angle of elevation of antenna beam is 20° . The earth radius can be assumed to be 6370km.
- Q. VII Discuss about the Ionospheric abnormalities in details
- Q. VIII With reference to the sky wave propagation explain the following terms:
- a. virtual height
 - b. Critical frequency.
 - c. Skip distance.
 - d. Maximum Usable Frequency

